

Exercise 2.2: Let Z be an $n \times r$ null-space matrix for the matrix A . If Y is any invertible $r \times r$ matrix, prove that $\hat{Z} = ZY$ is also a null space matrix for A .

Proof. Suppose Z is an $n \times r$ null-space matrix for the matrix A and let Y be an invertible $r \times r$ matrix. Consider the matrix ZY and note that since Y is invertible, its columns form a basis in $\mathbb{R}^r$, thus for every $x \in \mathbb{R}^r$ we know that there exists a $y \in \mathbb{R}^r$ of the form $y = Y^{-1}x$. Therefore by substitution we know $ZYy = ZY Y^{-1}x = Zx$. Thus $N(A) = R(ZY)$. $\square$

Exercise 3.1: For each of the following matrices, compute a basis for the null space using variable reduction (with A written in the form (B, N)).

(i)

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \\ 0 & 1 & 0 & 1 \end{pmatrix}.$$

Solution. Writing A in form of (B, N) we can see that the first three columns are sufficient to form a basis in the output space, since they are linearly independent. This gives us that the last column $(1, 1, 1)^T$ forms the null space of A . Solving for a basis of the null space Z using the variable reduction method involves computing the following,

$$Z = \begin{pmatrix} -B^{-1}N \\ I \end{pmatrix}.$$

So we get,

$$Z = \begin{pmatrix} -B^{-1}N \\ I \end{pmatrix} = \begin{pmatrix} -\begin{pmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \\ \frac{1}{2} & -\frac{1}{2} & -1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \\ I \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ 1 \\ 1 \end{pmatrix}$$

There is a paragraph in the book stating we should likely never form Z explicitly because of this matrix inverse calculation. $\square$

(ii) Consider the following

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \end{pmatrix}.$$

Solution. Again writing A in the form of (B, N) we can see that any column will form a sufficient basis in the output space of A . We will select the first column for B , therefore the matrix $(1, 1, 1)$ forms the null space of A . Again solving for the null space basis we get,

$$Z = \begin{pmatrix} -B^{-1}N \\ I \end{pmatrix} = \begin{pmatrix} -(1)(1, 1, 1) \\ I_3 \end{pmatrix} \begin{pmatrix} -1 & -1 & -1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

□

(iii) Consider the following,

$$A = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

Proof. Solution Again we can let the first two columns form B and let the last two form N . Solving for the null space basis, Z we get

$$Z = \begin{pmatrix} -B^{-1}N \\ I \end{pmatrix} = \begin{pmatrix} -\begin{pmatrix} \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & -\frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix} \\ I_2 \end{pmatrix} = \begin{pmatrix} 0 & -1 \\ -1 & 0 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

□

Exercise 3.3: Consider the system $Ap = 0$, where

$$A = \begin{pmatrix} 1 & 2 & 0 & 2 \\ 2 & 1 & 2 & 4 \end{pmatrix}$$

Compute a basis for the null-space matrix of A using p_2 and p_3 as the basic variables. Use this to write a general expression for all solutions to this system. Could you do the same if p_1 and p_4 were basic variable?

Proof. Solution Using variable reduction with p_2 and p_3 as the basic variables we form a B and N ,

$$B = \begin{pmatrix} 2 & 0 \\ 1 & 2 \end{pmatrix}$$

$$N = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$$

Computing the basis for the null space of A , Z . Recall that since we are using p_2 and p_3 as basic variables we must reorder the rows of Z to match accordingly. Doing so we get,

$$Z = \begin{pmatrix} (1, 0) \\ -B^{-1}N \\ (0, 1) \end{pmatrix} = \begin{pmatrix} (1, 0) \\ -\begin{pmatrix} \frac{1}{2} & 0 \\ -\frac{1}{4} & \frac{1}{2} \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix} \\ (0, 1) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2} & -1 \\ -\frac{3}{4} & -\frac{3}{2} \\ 0 & 1 \end{pmatrix}$$

Since Z forms a basis for the null space of A , every p satisfying $Ap = 0$ will be of the form,

$$Zp_n = \begin{pmatrix} 1 & 0 \\ -\frac{1}{2} & -1 \\ -\frac{3}{4} & -\frac{3}{2} \\ 0 & 1 \end{pmatrix} \begin{pmatrix} p_1 \\ p_4 \end{pmatrix} = \begin{pmatrix} p_1 \\ -\frac{1}{2}p_1 - p_4 \\ -\frac{3}{4}p_1 - \frac{3}{2}p_4 \\ p_4 \end{pmatrix}.$$

Finally suppose we had chosen p_1 and p_4 to be our basic variables. Note that columns corresponding to p_1 and p_4 , have the property that $2\hat{p}_1 = \hat{p}_4$ thus they are linearly dependent. Forming B will result in a singular matrix, so we cannot compute B^{-1} . Also note the method for finding p without forming Z involves solving this system $Bu = -Np_n$ which is equally problematic when B is singular. $\square$

Exercise 3.4: Let A be an $m \times n$ matrix of full row rank. Prove that the matrix AA^T is positive definite, and hence its inverse exists.

Proof. Suppose that A is an $m \times n$ matrix of full row rank. Let $A = U\Sigma V^T$ where U is an $m \times m$ unitary matrix and V^T is an $n \times n$ unitary matrix and Σ is an $m \times n$ diagonal matrix of singular values. Since A is full row rank, we know that Σ must also be full row rank. Now consider that by substitution,

$$AA^T = U\Sigma V^T (U\Sigma V^T)^T = (U\Sigma V^T)(V\Sigma^T U^T)$$

Since V is unitary and Σ is diagonal we get the following,

$$AA^T = U(\Sigma\Sigma^T)U^T = U(\Sigma^2)U^T$$

Note that U and U^T are $m \times m$ and $\Sigma\Sigma^T$ is a diagonal $m \times m$ matrix. Recall that the eigenvalues of AA^T can be found by square rooting the corresponding singular values of AA^T . Since Σ was full row rank we know that $\Sigma\Sigma^T = \Sigma^2$ has all positive non-zero entries on the diagonal. Therefore $\Lambda = \sqrt{\Sigma^2} = \Sigma$ and the eigenvalues of AA^T are all positive non-zero. Thus AA^T is positive definite and $(AA^T)^{-1} = U^T(1/\Sigma^2)U$. $\square$

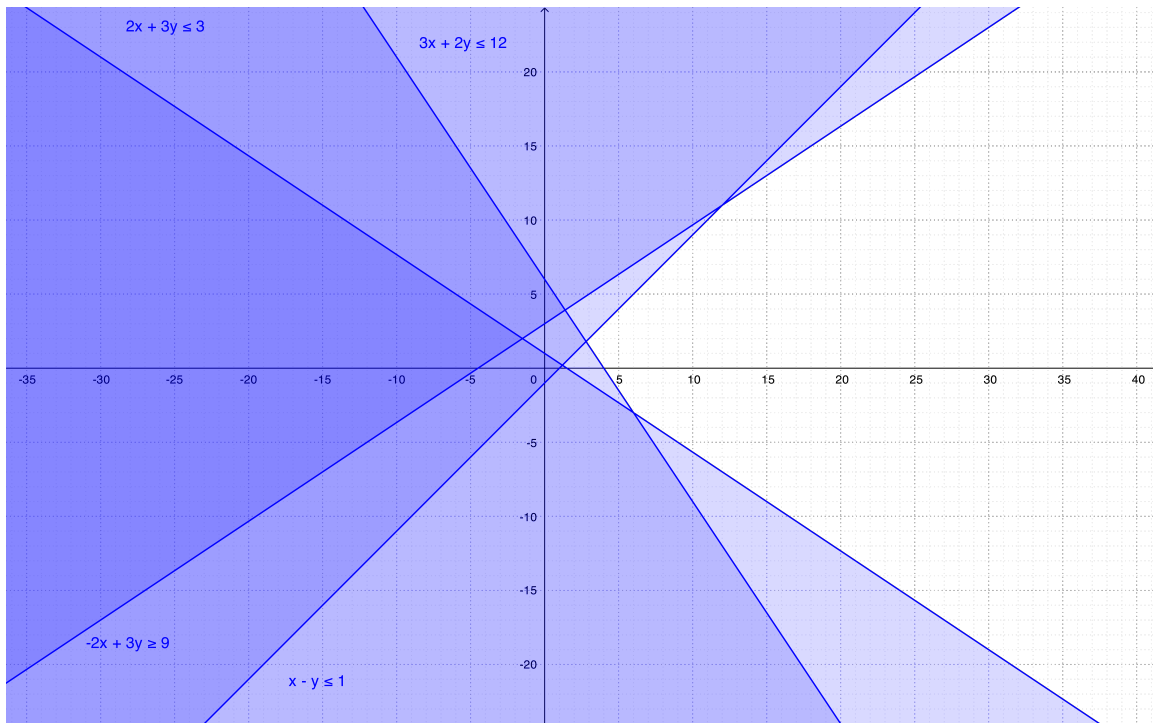
Exercise 1.1: Solve the following linear programs graphically,

(i)

Minimize: $z = 3x + y$ Subject to: $x - y \leq 1$ $3x + 2y \leq 12$ $2x + 3y \leq 3$ $-2x + 3y \geq 9$ $x, y \geq 0$

Solution. Plotting the first four inequality constraints we find that there the feasible set must be empty. Every solution to the first four inequality constraints falls outside the first quadrant where $x, y \geq 0$. Thus this problem has no solution. (Empty feasible set)

Figure 1: First four inequality constraints. (Plotted with GeoGebra)

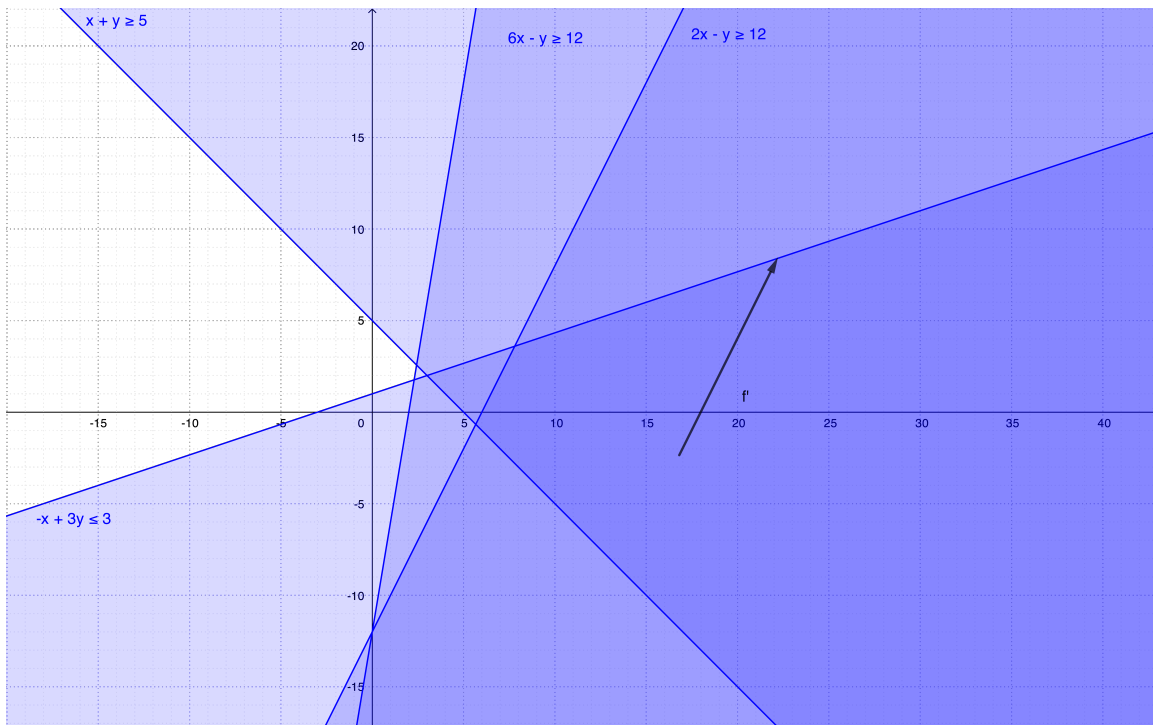


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(ii)

Maximize: $z = x + 2y$ Subject to: $2x - y \geq 12$ $x + y \geq 5$ $-x + 3y \leq 3$ $6x - y \geq 12$ $x, y \geq 0$

Solution. Plotting the first four inequality constraints we see that there feasible set is unbounded. Finding our ascent direction $\nabla z = (1, 2)^T$. Plotting our direction vector we see that it points towards the $-x + 3y \leq 3$ inequality constraint in a direction of unboundedness. Thus there is no maximum.

Figure 2: First four inequality constraints with ascent direction f' . (Plotted with GeoGebra)

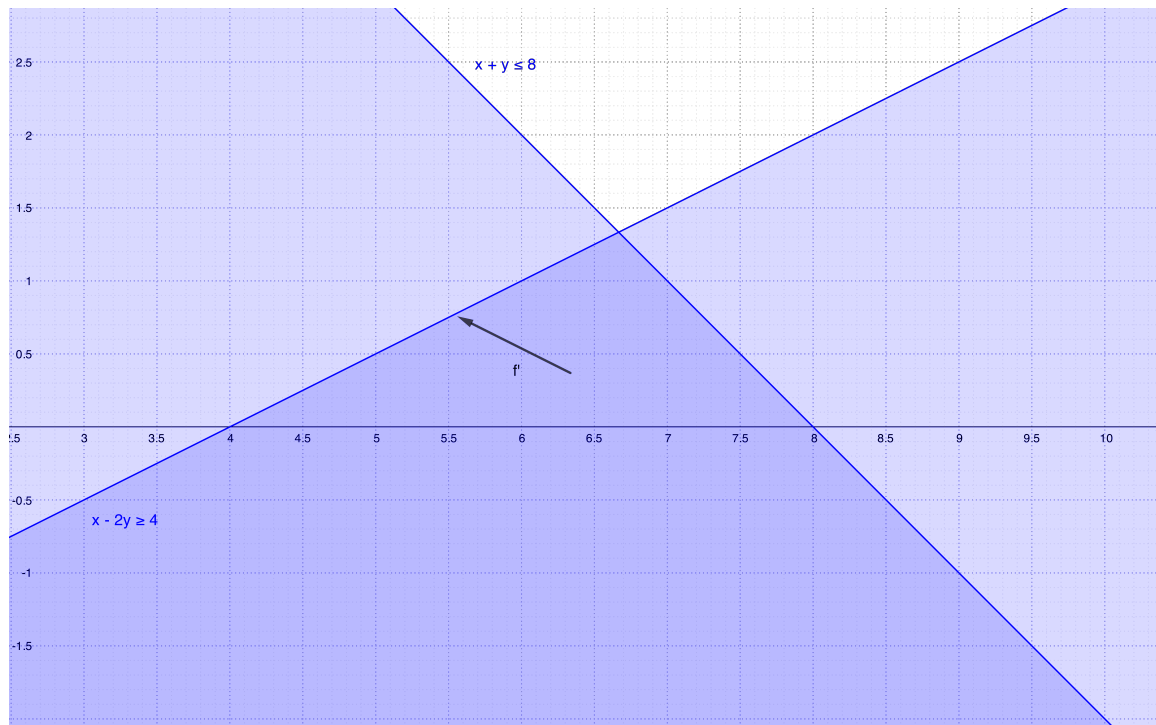
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(iii)

Minimize: $z = x - 2y$ Subject to: $x - 2y \geq 4$ $x + y \leq 8$ $x, y \geq 0$

Solution. Plotting the first two inequality constraints we see that there feasible set is bounded. Finding our descent direction $-\nabla z = (-1, 2)^T$. Finally we see that our descent direction is pointing towards the $x - 2y \leq 4$ inequality so anywhere on the line $x - 2y = 4$ where we maintain feasible is a solution to this linear program.

Figure 3: First two inequality constraints with descent direction f' . (Plotted with GeoGebra)



□

(iv)

Minimize: $z = -x - y$

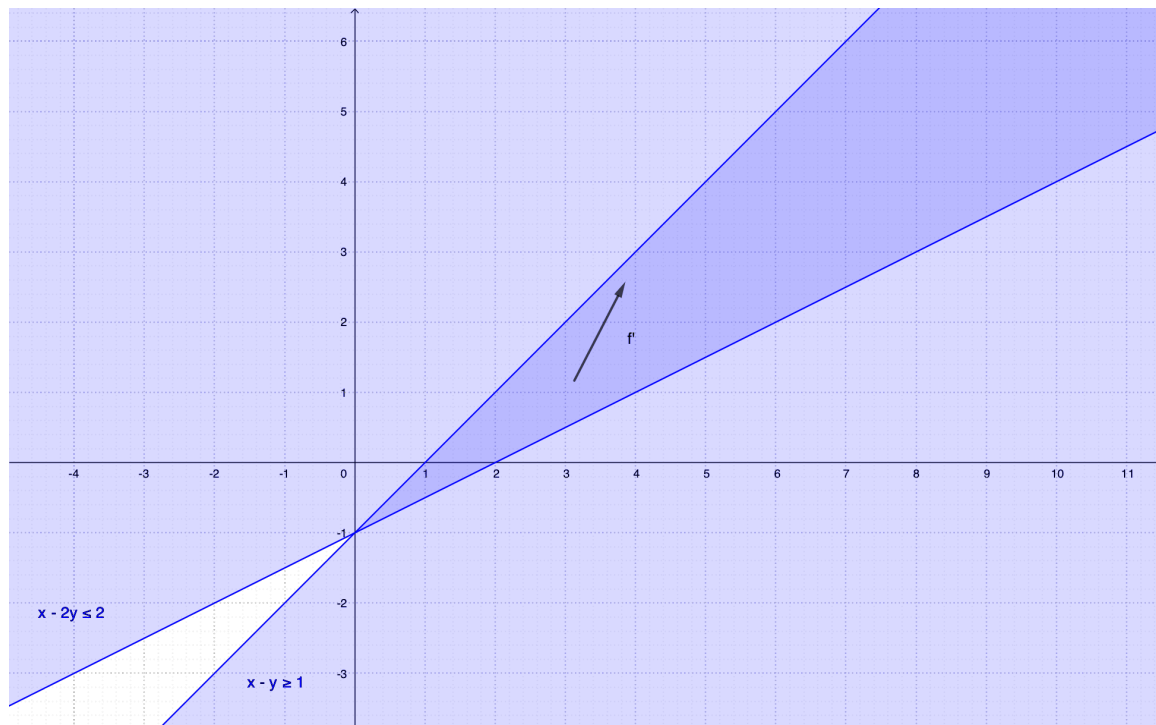
Subject to: $x - y \geq 1$

$x - 2y \leq 2$

$x, y \geq 0$

Solution. Plotting the first two inequality constraints we see that there feasible set is bounded. Finding our descent direction $-\nabla z = (1, 1)^T$. Finally we see that our descent direction is pointing towards the $x - 2y \leq 2$ inequality and in a direction of unboundedness. Thus there is no minimum.

Figure 4: First two inequality constraints with descent direction f' . (Plotted with GeoGebra)



□

Exercise 1.2: Find graphically all the values of the parameter a such that $(-3, 4)^T$ is the optimal solution of the following problem:

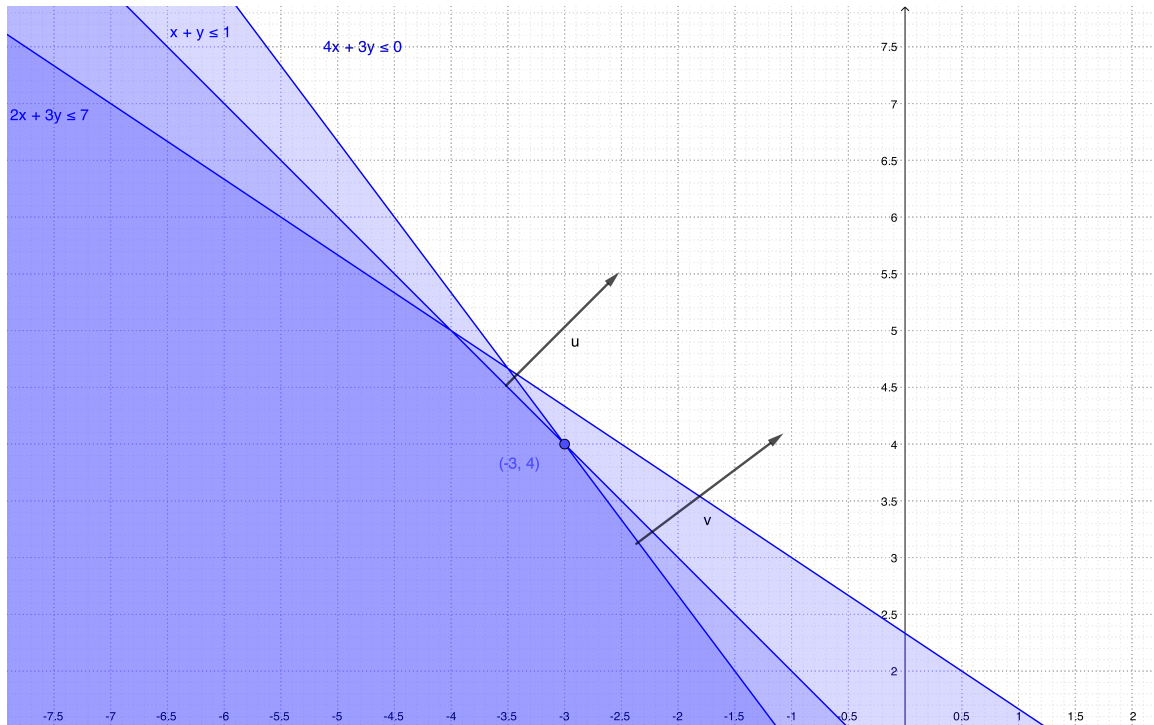
$$\text{Maximize: } z = ax + (2 - a)y$$

$$\text{Subject to: } 4x + 3y \leq 0$$

$$2x + 3y \leq 7$$

$$x + y \leq 1$$

Looking at the plot of the feasible set, for $(-3, 4)^T$ to be the optimal solution our ascent direction $\nabla z = (a, 2 - a)^T$ must be a convex combination of the two vectors orthogonal to the inequalities that produce the intersection at $(-3, 4)^T$, namely $u = (1, 1)^T$ and $v = (4, 3)^T$.

Figure 5: Plotting the feasible set with orthogonal vectors u and v . (Plotted with GeoGebra)

Thus $(-3, 4)^T$ is an optimal solution for all ∇z where for all $0 \leq \alpha \leq 1$,

$$\nabla z = \begin{pmatrix} a \\ 2 - a \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 1 \end{pmatrix} + (1 - \alpha) \begin{pmatrix} 4 \\ 3 \end{pmatrix}.$$

Note that we didn't use strict inequalities for α since $(-3, 4)^T$ would still be an optimal solution in those cases, just not the only one.